



1. An array of 2 two byte integers is stored in big endian machine in byte addresses as shown below. What will be its storage pattern in little endian machine?

Address	Data
0×104	78
0×103	56
0×102	34
0×101	12

(a) 0×104 12
 0×103 56
 0×102 34
 0×101 78

(b) 0×104 12
 0×103 34
 0×102 56
 0×101 78

(c) 0×104 56
 0×103 78
 0×102 12
 0×101 34

(d) 0×104 56
 0×103 12
 0×102 78
 0×101 34

2. A non-pipelined CPU has 12 general purpose registers (R0, R1, R2,.....R12). Following operations are supported

ADD Ra, Rb, Rr Add Ra to Rb and store the result in Rr

MUL Ra, Rb, Rr Multiply Ra to Rb and store the result in Rr

MUL operations takes two clock cycles, ADD takes one clock cycle.

Calculate minimum number of clock cycles required to compute the value of the expression $XY + XYZ + YZ$. The variables X, Y, Z are initially available in registers R0, R1 and R2 and contents of these registers must not be modified.

(a) 5
(c) 7

(b) 6
(d) 8



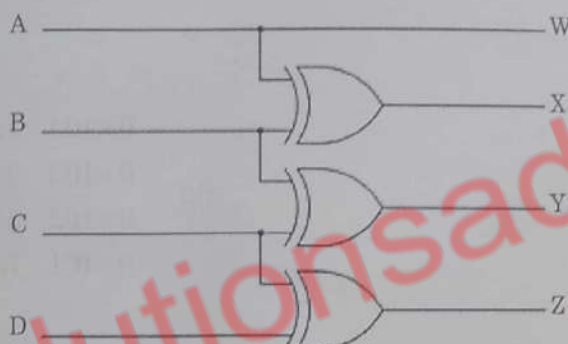
3. Minimum number of NAND gates required to implement the following binary equation

$$Y = (\overline{A} + \overline{B})(C + D)$$

- (a) 4
(c) 3

- (b) 5
(d) 6

4. IF ABCD is a 4-bit binary number, then what is code generated by the following circuit?



- (a) BCD code
(c) 8421 code

- (b) Gray code
(d) Excess-3 code

5. The number of tokens in the following C code segment is

```
switch(inputvalue)
{
case 1 : b = c * d; break;
default : b = b++; break;
}
```

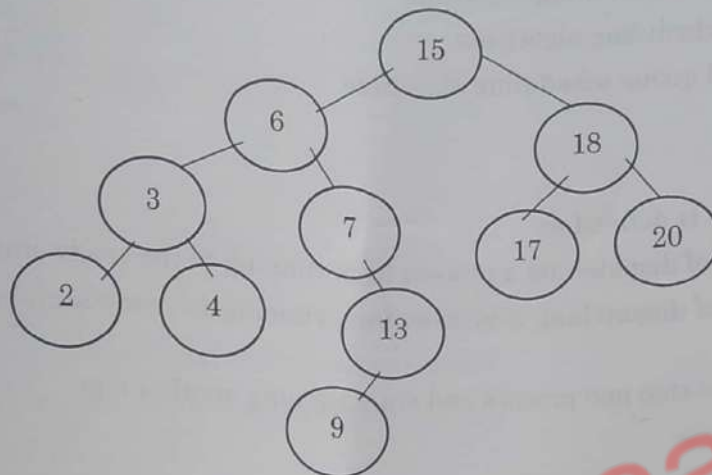
- (a) 27
(c) 26

- (b) 29
(d) 24

6. In a two-pass assembler, resolution of subroutine calls and inclusion of labels in the symbol table is done during

- (a) second pass
(b) first pass and second pass respectively
(c) second pass and first pass respectively
(d) first pass

7. What is the in-order successor of 15 in the given binary search tree?



- (a) 18
(c) 17

- (b) 6
(d) 20

8. The minimum height of an AVL tree with n nodes is

(a) $\text{Ceil}(\log_2(n+1))$

(b) $1.44 \log_2 n$

(c) $\text{Floor}(\log_2(n+1))$

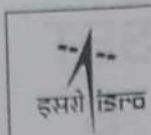
(d) $1.64 \log_2 n$

9. The master theorem

- (a) assumes the subproblems are unequal sizes
(b) can be used if the subproblems are of equal size
(c) cannot be used for divide and conquer algorithms
(d) cannot be used for asymptotic complexity analysis

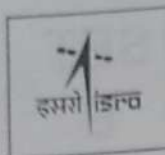
10. Raymonds tree based algorithm ensures

- (a) no starvation, but deadlock may occur in rare cases
(b) no deadlock, but starvation may occur
(c) neither deadlock nor starvation can occur
(d) deadlock may occur in cases where the process is already starved



11. Which of the following algorithms defines time quantum?
- (a) shortest job scheduling algorithm
 - (b)/ round robin scheduling algorithm
 - (c) priority scheduling algorithm
 - (d) multilevel queue scheduling algorithm
12. Dispatch latency is defined as
- (a) the speed of dispatching a process from running to the ready state
 - (b) the time of dispatching a process from running to ready state and keeping the CPU idle
 - (c) the time to stop one process and start running another one
 - (d) none of these
13. An aid to determine the deadlock occurrence is
- (a)/ resource allocation graph
 - (b) starvation graph
 - (c) inversion graph
 - (d) none of the above
14. Consider the following page reference string.
1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6
What are the minimum number of frames required to get a single page fault for the above sequence assuming LRU replacement strategy?
- (a) 7
 - (b) 4
 - (c)/ 6
 - (d) 5
15. Three CPU-bound tasks, with execution times of 15, 12 and 5 time units respectively arrive at times 0, t and 8, respectively. If the operating system implements a shortest remaining time first scheduling algorithm, what should be the value of t to have 4 context switches? Ignore the context switches at time 0 and at the end.
- (a) $0 < t < 3$
 - (b) $t = 0$
 - (c) $t \leq 3$
 - (d) $3 < t < 8$

16. Context free languages are closed under
- union, intersection
 - union, kleene closure
 - intersection, complement
 - complement, kleene closure ✓
17. Which of the following is true?
- Every subset of a regular set is regular/
 - Every finite subset of non-regular set is regular
 - The union of two non regular set is not regular
 - Infinite union of finite set is regular
18. The language which is generated by the grammar $S \rightarrow aSa \mid bSb \mid a \mid b$ over the alphabet $\{a, b\}$ is the set of
- Strings that begin and end with the same symbol
 - All odd and even length palindromes ✗
 - All odd length palindromes ✓
 - All even length palindromes ✗
19. Which of the following classes of languages can validate an IPv4 address in dotted decimal format? It is to be ensured that the decimal values lie between 0 and 255.
- RE and higher ✓
 - CFG and higher
 - CSG and higher
 - Recursively enumerable language
20. Minimum number of states required in DFA accepting binary strings not ending in "101" is
- 3
 - (b) 4 ✓
 - (c) 5
 - (d) 6
21. Which of the following is a type of a out-of-order execution, with the reordering done by a compiler
- loop unrolling
 - dead code elimination
 - strength reduction
 - software pipelining



22. One instruction tries to write an operand before it is written by previous instruction. This may lead to a dependency called
- (a) True dependency
 - (b) Anti-dependency
 - (c) Output Dependency
 - (d) Control hazard
23. If every non-key attribute functionally dependant on the primary key, then the relation will be in
- (a) First normal form
 - (b) Second normal form
 - (c) Third normal form
 - (d) Fourth normal form
24. The SQL query
- ```
SELECT columns
FROM TableA
RIGHT OUTER JOIN TableB
ON A.columnName = B.columnName
WHERE A.columnName IS NULL
```
- returns the following :
- (a) All rows in TableB, which meets equality condition above and, none from Table A which meets the condition
  - (b) All rows in TableA, which meets equality condition above and none from Table B, which meets the condition
  - (c) All rows in TableB, which meets equality condition
  - (d) All rows in TableA, which meets equality condition
25. To send same bit sequence, NRZ encoding require
- (a) Same clock frequency as Manchester encoding
  - (b) Half the clock frequency as Manchester encoding
  - (c) Twice the clock frequency as Manchester encoding
  - (d) A clock frequency which depend on number of zeros and ones in the bit sequence
26. The persist timer is used in TCP to
- (a) To detect crashes from the other end of the connection
  - (b) To enable retransmission
  - (c) To avoid deadlock condition
  - (d) To timeout FIN\_Wait1 condition



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27. Remote Procedure Calls are used for
- (a) communication between two processes remotely different from each other on the same system ✓
  - (b) communication between two processes on the same system
  - (c) communication between two processes on separate systems
  - (d) none of the above

28. Consider the following recursive C function that takes two arguments  
unsigned int rer (unsigned int n, unsigned int r) {  
if (n > 0) return (n%r + rer (n/r, r));  
else return 0;  
}

What is the return value of the function rer when it is called as rer (513, 2)?

- (a) 9                      (b) 8                      (c) 5                      (d) 2 ✓

29. A given grammar is called ambiguous if
- (a) two or more productions have the same non-terminal on the left hand side
  - (b) a derivation tree has more than one associated sentence
  - (c) there is a sentence with more than one derivation tree corresponding to it ✓
  - (d) brackets are not present in the grammar

30. What is the output of the code given below?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
char name[]="satellites";
```

```
int len;
```

```
int size;
```

```
len = strlen(name);
```

```
size = sizeof(name);
```

```
printf("%d" , len * size);
```

```
return 0;
```

```
}
```

(a) 100

(b) 110 ✓

(c) 40

(d) 44





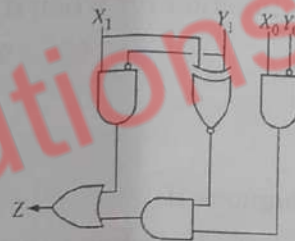
31. Regression testing is primarily related to

- (a) Functional testing/
- (b) Development testing
- (c) Data flow testing
- (d) Maintenance testing

32. Of the following sort algorithms, which has execution time that is least dependant on initial ordering of the input?

- (a) Insertion sort
- (b) Quick sort
- (c) Merge sort/
- (d) Selection sort

33. The following circuit compares two 2-bit binary numbers,  $X$  and  $Y$  represented by  $X_1X_0$  and  $Y_1Y_0$  respectively. ( $X_0$  and  $Y_0$  represent Least Significant Bits)



Under what condition  $Z$  will be 1?

- (a)  $X > Y$
- (b)  $X < Y$
- (c)  $X = Y$
- (d)  $X \neq Y$

34. What is the availability of the software with following reliability figures

Mean Time Between Failures (MTBF) is 20 days

Mean Time To Repair (MTTR) is 20 hours

- (a) 90%
- (b) 96%
- (c) 24% ✓
- (d) 50%

35. What is the defect rate for Six sigma?

- (a) 1.0 defect per million lines of code
- (b) 1.4 defects per million lines of code
- (c) 3.0 defects per million lines of code
- (d) 3.4 defects per million lines of code



36. A stack is implemented with an array of 'A [0..N-1]' and a variable 'pos'. The push and pop operations are defined by the following code.

```

push (x)
 A[pos] ← x
 pos ← pos - 1
end push
pop ()
 pos ← pos + 1
 return A[pos]
end pop

```

Which of the following will initialize an empty stack with capacity  $N$  for the above implementation?

- (a)  $pos \leftarrow -1$       (b)  $pos \leftarrow 0$       (c)  $pos \leftarrow 1$       (d)  $pos \leftarrow N-1$

37. Given that

$B(a)$  means " $a$  is a bear"

$F(a)$  means " $a$  is a fish" and

$E(a, b)$  means " $a$  eats  $b$ "

Then what is the best meaning of

$\forall x [F(x) \rightarrow \exists y (E(y, x) \rightarrow b(y))]$

- (a) Every fish is eaten by some bear      (b) Bears eat only fish  
(c) Every bear eats fish/      (d) Only bears eat fish

38. Following declaration of an array of struct, assumes size of byte, short, int and long are 1, 2, 3 and 4 respectively. Alignment rule stipulates that  $n$ -byte field must be located at an address divisible by  $n$ . the fields in a struct are not rearranged, padding is used to ensure alignment. All elements of array should be of same size.

Struct complx

|         |   |   |
|---------|---|---|
| Short s | 2 | 8 |
| Byte b  | 1 | 8 |
| Long l  | 4 | 8 |
| Int i   | 3 | 8 |

End complx

Complx C[10]

Assuming  $C$  is located at an address divisible by 8, what is the total size of  $C$ , in bytes?

- (a) 150      (b) 160/      (c) 200      (d) 240



39. In the following procedure  
Integer procedure P(X,Y);  
Integer X,Y;  
value x;  
begin

K = 5;  
L = 8;  
P = x + y;

end

X is called by value and Y is called by name. If the procedure were invoked by the following program fragment

K = 0;  
L = 0;  
Z = P(K,L);

then the value Z would be set equal to

- (a) 5 (b) 8 ✓  
(c) 13 (d) 0

40. Consider product of three matrices  $M_1, M_2$  and  $M_3$  having  $w$  rows and  $x$  columns,  $x$  rows and  $y$  columns, and  $y$  rows and  $z$  columns. Under what condition will it take less time to compute the product as  $(M_1 M_2) M_3$  than to compute  $M_1 (M_2 M_3)$ ?

- (a) Always take the same time (b)  $(1/x + 1/z) < (1/w + 1/y)$   
(c)  $x > y$  (d)  $(w + x) > (y + z)$

41. A new flipflop with inputs  $X$  and  $Y$ , has the following property

| Inputs |     | Current state | Next state     |
|--------|-----|---------------|----------------|
| $X$    | $Y$ |               |                |
| 0      | 0   | $Q$           | 1              |
| 0      | 1   | $Q$           | $\overline{Q}$ |
| 1      | 1   | $Q$           | 0              |
| 1      | 0   | $Q$           | $Q$            |

Which of the following expresses the next state in terms of  $X, Y$ , current state?

- (a)  $(\overline{X} \wedge \overline{Q}) \vee (\overline{Y} \wedge Q)$  (b)  $(\overline{X} \wedge Q) \vee (\overline{Y} \wedge \overline{Q})$   
(c)  $(X \wedge \overline{Q}) \vee (Y \wedge Q)$  (d)  $(X \wedge \overline{Q}) \vee (\overline{Y} \wedge Q)$

42. The immediate addressing mode can be used for
1. Loading internal registers with initial values
  2. Perform arithmetic or logical operation on data contained in instructions
- Which of the following is true?

- (a) Only 1  
(b) Only 2  
(c) Both 1 and 2  
(d) Immediate mode refers to data in cache

43. Statements associated with registers of a CPU are given. Identify the false statement.
- (a) The program counter holds the memory address of the instruction in execution.
  - (b) Only opcode is transferred to the control unit
  - (c) An instruction in the instruction register consists of the opcode and the operand
  - (d) The value of the program counter is incremented by 1 once its value has been read to the memory address register

44. Which of the following affects the processing power assuming they do not influence each other.

1. Data bus capability
2. Addressing scheme
3. Clock speed

- (a) 3 only  
(b) 1 and 3 only  
(c) 2 and 3 only  
(d) 1, 2 and 3

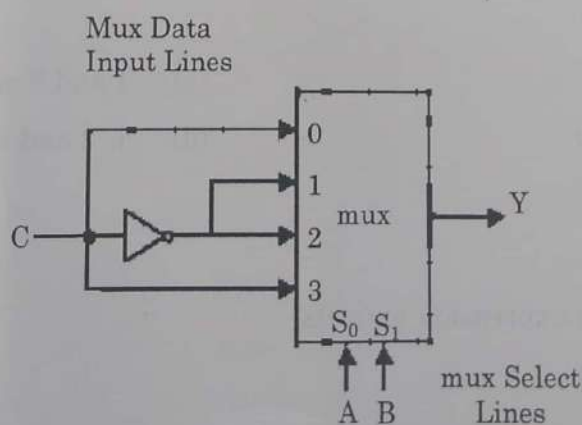
45. Convert the pre-fix expression to in-fix

$- * + ABC * - DE + FG$

- (a)  $(A-B)*C+(D*E)-(F+G)$
- (b)  $(A+B)*C-(D-E)*(F-G)$
- (c)  $(A+B-C)*(D-E)*(F+G)$
- (d)  $(A+B)*C-(D*E)-(F+G)$



46. Consider a 5-segment pipeline with a clock cycle time 20ns in each sub operation. Find out the approximate speed-up ratio between pipelined and non-pipelined system to execute 100 instructions. (if an average, every five cycles, a bubble due to data hazard has to be introduced in the pipeline)
- (a) 5 (b) 4.03  
(c) 4.81 (d) 4.17
47. Consider a 32-bit processor which supports 70 instructions. Each instruction is 32 bit long and has 4 fields namely opcode, two register identifiers and an immediate operand of unsigned integer type. Maximum value of the immediate operand that can be supported by the processor is 8191. How many registers the processor has?
- (a) 32 (b) 64  
(c) 128 (d) 16
48. In a 8-bit ripple carry adder using identical full adders, each full adder takes 34 ns for computing sum. If the time taken for 8-bit addition is 90 ns, find time taken by each full adder to find carry.
- (a) 6 ns (b) 7 ns  
(c) 10 ns (d) 8 ns
49. Following Multiplexer circuit is equivalent to



- (a) Sum equation of full adder  
(b) Carry equation of full adder  
(c) Borrow equation for full subtractor  
(d) Difference equation of a full subtractor

50. A stack organised computer is characterised by instructions with
- (a) indirect addressing (b) direct addressing
- (c) zero addressing (d) index addressing
51. A computer which issues instructions in order, has only 2 registers and 3 opcodes ADD, SUB and MOV. Consider 2 different implementations of the following basic block :

| Case 1          | Case 2          |
|-----------------|-----------------|
| $t1 = a + b;$   | $t2 = c + d;$   |
| $t2 = c + d;$   | $t3 = e - t2;$  |
| $t3 = e - t2;$  | $t1 = a + b;$   |
| $t4 = t1 - t2;$ | $t4 = t1 - t2;$ |

Assume that all operands are initially in memory. Final value of computation also has to reside in memory. Which one is better in terms of memory accesses and by how many MOV instructions?

- (a) Case 2, 2 (b) Case 2, 3
- (c) Case 1, 2 (d) Case 1, 3
52. Which one indicates a technique of building cross compilers?
- (a) Beta cross (b) Canadian cross
- (c) Mexican cross (d) X-cross

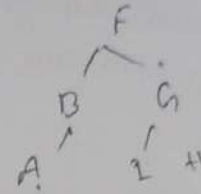
53. Consider a 2-dimensional array  $x$  with 10 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in row-major format. If the first element  $x[0][0]$  occupies the memory location with address 1000 and each element occupies only one memory location, which all locations (in decimal) will be holding a value of 10?

- (a) 1018, 1019 (b) 1022, 1041
- (c) 1013, 1014 (d) 1000, 1399



54. The post-order traversal of a binary tree is ACEDBHIGF. The pre-order traversal is

- (a) ABCDEFGHI
- (b) FBADCEGIH
- (c) FABCDEGHI
- (d) ABDCEFGIH



55. In linear hashing, if blocking factor bfr, loading factor  $i$  and file buckets  $N$  are known, the number of records will be

- (a)  $cr = i + bfr + N$
- (c)  $r = i + bfr - N$

(b)  $r = i - bfr - N$

(d)  $r = i * bfr * N$

56. What is compaction refers to

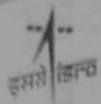
- (a) a technique for overcoming internal fragmentation ✓
- (b) a paging technique
- (c) a technique for overcoming external fragmentation
- (d) a technique for compressing the data

57. The operating system and the other processes are protected from being modified by an already running process because

- (a) they run at different time instants and not in parallel
- (b) they are in different logical addresses
- (c) they use a protection algorithm in the scheduler
- (d) every address generated by the CPU is being checked against the relocation and limit parameters ✓



58. G is an undirected graph with vertex set  $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$  and edge set  $\{v_1v_2, v_1v_3, v_1v_4, v_2v_4, v_2v_5, v_3v_4, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$ . A breadth first search of the graph is performed with  $v_1$  as the root node. Which of the following is a tree edge?
- (a)  $v_2v_4$  (b)  $v_1v_4$  (c)  $v_4v_5$  (d)  $v_3v_4$
59. If the array A contains the items 10, 4, 7, 23, 67, 12 and 5 in that order, what will be the resultant array A after third pass of insertion sort?
- (a) 67, 12, 10, 5, 4, 7, 23 (b) 4, 7, 10, 23, 67, 12, 5 ✓  
(c) 4, 5, 7, 67, 10, 12, 23 (d) 10, 7, 4, 67, 23, 12, 5
60. Huffman tree is constructed for the following data : {A, B, C, D, E} with frequency {0.17, 0.11, 0.24, 0.33 and 0.15} respectively. 100 00 01101 is decoded as
- (a) BACE (b) CADE (c) BAD (d) CADD
61. Given the grammar
- $$s \rightarrow T * S \mid T$$
- $$T \rightarrow U + T \mid U$$
- $$U \rightarrow a \mid b$$
- Which of the following statements is wrong?
- (a) Grammar is not ambiguous  
(b) Priority of + over \* is ensured  
(c) Right to left evaluation of \* and + happens ✓  
(d) None of these
62. What is the complexity of the following code?
- ```
sum=0;
for (i=1; i <= n; i*=2)
    for(j=1; j<=n;j++)
        sum++;
```
- Which of the following is not a valid string?
- (a) $O(n^2)$ (b) $O(n \log n)$
(c) $O(n)$ (d) $O(n \log n \log n)$



63. Which of the following is an efficient method of cache updating?
- (a) Snoopy writes
 - (b) Write through
 - (c) Write within
 - (d) Buffered write
64. In a columnar transposition cipher, the plain text is "the tomato is a plant in the night shade family", keyword is "TOMATO". The cipher text is
- the tomato is a plant in the night shade*
- (a) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"
 - (b) "TINESAX / EOAHTFX / MAIIAIX / HTLTHEY / TAPNGDL / OSTNHMX"
 - (c) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / OSTNHMX / TAPNGDL"
 - (d) "EOAHTFX / TINESAX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"
65. Avalanche effect in cryptography refers
- (a) Large changes in cipher text when the keyword is changed minimally
 - (b) Large changes in cipher text when the plain text is changed
 - (c) Large Impact of keyword change to length of the cipher text
 - (d) None of the above
66. A magnetic disk has 100 cylinders, each with 10 tracks of 10 sectors. If each sector contains 128 bytes, what is the maximum capacity of the disk in kilobytes?
- (a) 1,280,000
 - (b) 1280
 - (c) 1250
 - (d) 128,000
67. How many total bits are required for a direct-mapped cache with 128 KB of data and 1 word block size, assuming a 32-bit address and 1 word size of 4 bytes?
- (a) 2 Mbits
 - (b) 1.7 Mbits
 - (c) 2.5 Mbits
 - (d) 1.5 Mbits
68. Properties of 'DELETE' and 'TRUNCATE' commands indicate that
- (a) After the execution of 'TRUNCATE' operation, COMMIT, and ROLLBACK statements cannot be performed to retrieve the lost data, while 'DELETE' allow it
 - (b) After the execution of 'DELETE' and 'TRUNCATE' operation retrieval is easily possible for the lost data
 - (c) After the execution of 'DELETE' operation, COMMIT and ROLLBACK statements can be performed to retrieve the lost data., while TRUNCATE do not allow it
 - (d) After the execution of 'DELETE' and 'TRUNCATE' operation no retrieval is possible for the lost data

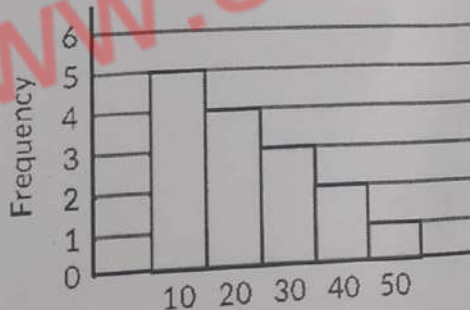
69. Checksum field in TCP header is
- ones complement of sum of header and data in bytes
 - ones complement of sum of header, data and pseudo header in 16 bit words
 - dropped from IPv6 header format
 - better than md5 or sh1 methods

70. If $x + 2y = 30$, then
- $$\left(\frac{2y}{5} + \frac{x}{3}\right) + \left(\frac{x}{5} + \frac{2y}{3}\right)$$
- will be equal to

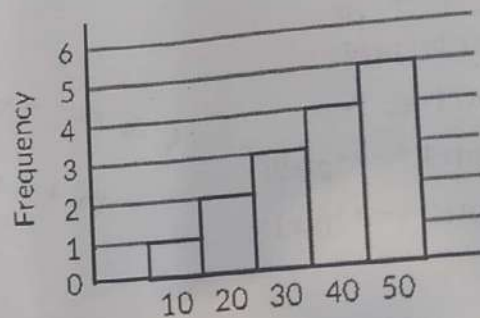
- 8
- 16
- 18
- 20

71. For the distributions given below :

Distribution A



Distribution B



Which of the following is correct for the above distributions?

- Standard deviation of A is significantly lower than standard deviation of B
- Standard deviation of A is slightly lower than standard deviation of B
- Standard deviation of A is same as standard deviation of B
- Standard deviation of A is significantly higher than standard deviation of B

72. The hardware implementation which provides mutual exclusion is

- Semaphores
- Test and set instruction ✓
- Both options
- None of the options



73. What is the output of the following 'c' code assuming it runs on a byte addressed little endian machine?

```
#include <stdio.h>
int main()
{
    int x; char *ptr;
    x = 622, 100, 101;
    printf("%d", (*(char *)&x) * (x % 3));
    return 0;
}
```

(a) 622

(b) 311

(c) 22

(d) 110

74. What is the output in a 32 bit machine with 32 bit compiler?

```
#include <stdio.h>
rer(int **ptr2, int **ptr1)
{
    int *ii;
    ii = *ptr2;
    *ptr2 = *ptr1;
    *ptr1 = ii;
    **ptr1 *= **ptr2;
    **ptr2 += **ptr1;
}
```

void main()

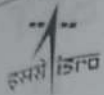
```
{
    int var1=5, var2=10;
    int *ptr1=&var1, *ptr2=&var2;
    rer(&ptr1, &ptr2);
    printf("%d %d", var2, var1);
}
```

(a) 60 70

(c) 50 60

(b) 50 50

(d) 60 50



75. Consider the following pseudo-code

$I=0; J=0; K=8;$

while ($I < K-1$) // while-1

{

$J=J+1;$

while ($J < K$) //while-2

{

if ($x[I] < x[J]$)

{

$temp = x[I];$

$x[I] = x[J];$

$x[J] = temp;$

}

} // end of while-2

$I=I+1;$

} // end of while-1

The cyclomatic complexity of the above is

(a) 3

(b) 2

(c) 4

(d) 1

76. In a class definition with 10 methods, to make the class maximally cohesive, number of direct and indirect connections required among the methods are

(a) 90, 0

(b) 45, 0

(c) 10, 10

(d) 45, 45

77. Of the following, which best approximates the ratio of the number of nonterminal nodes in the total number of nodes in a complete K -ary tree of depth N ?

(a) $1/N$

(b) $N-1/N$

(c) $1/K$

(d) $K-1/K$



78. A grammar is defined as

$$A \rightarrow BC$$

$$B \rightarrow x \mid Bx$$

$$C \rightarrow B \mid D$$

$$D \rightarrow y \mid Ey$$

$$E \rightarrow z$$

The non-terminal alphabet of the grammar is

(a) $\{A, B, C, D, E\}$

(c) $\{A, B, C, D, E, x, y, z\}$

(b) $\{B, C, D, E\}$

(d) $\{x, y, z\}$

79. If $A = \{x, y, z\}$ and $B = \{u, v, w, x\}$, and the universe is $\{s, t, u, v, w, x, y, z\}$
Then $(A \cup \bar{B}) \cap (A \cap B)$ is equal to

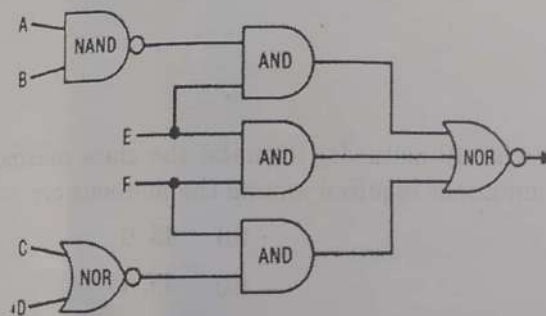
(a) $\{u, v, w, x\}$

(c) $\{u, v, w, x, y, z\}$

(b) $\{\}$

(d) $\{u, v, w\}$

80. Consider the following circuit



The function by the network above is

(a) $\overline{ABE} + EF + \overline{CDF}$

(c) $(\overline{AB} + E)(\overline{E} + \overline{F})(C + D + \overline{F})$

(b) $(\overline{E} + AB\overline{F})(C + D + \overline{F})$

(d) $(A + B)\overline{E} + \overline{EF} + C\overline{D}\overline{F}$